Meixi Chen

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Education and Qualifications

M.Eng. & B.A. Engineering, University of Cambridge *Oct. 2018 – Jul. 2023*

2022-2023: Master's degree with Merit.

- Class I in Project: *Automatic Children Story Video Generation*
  - Text-to-image generation as main research focus.
  - DreamBooth benchmark improved against prompt accuracy and subject consistency by manual control of the level of training via weight interpolation at inference stage.

2021-2022: Intermission year in industry; 2020-2021: Class II.1;
2019-2020: Not classed due to COVID; 2018-2019: Class II.1.

Course modules related to research interests:

- Deep learning and structured data, Computer vision, Practical optimisation, Probabilistic machine learning, Signals and systems, Inference.

Harrow International School Hong Kong *Sept. 2016 – Jun. 2018*

A-Levels (Edexcel): Mathematics (A*), Further Mathematics (A*), Physics (A*), Chemistry (A*).

Research Experiences

Automatic Children Story Video Generation

Supervised by Dr. Samuel Albanie

M.Eng. project, University of Cambridge (Oct. 2022 - Jul. 2023)

- Text-to-image diffusion models used to generate images for the video
  - Personalization of text-to-image diffusion models, in order to maintain character consistency: the model learns a new concept/object/character from user input few-shot training images.
  - DreamBooth method as benchmark: DreamBooth fine-tunes the diffusion model to achieve personalised images. It is typically difficult to tune, needing different fine-tuning steps for different text prompts.
  - Additional manual control to improve DreamBooth via weight interpolation at inference stage. By intentionally over-training the model then manually tune the training level by merging fine-tuned weights with the weights before fine-tuning, different training levels can be achieved according to different requirements, with no needs for further training.
- Automatic generation pipeline built: character design images to text description (fine-tuned ViT) + story generation (ChatGPT) + image generation (text prompts by ChatGPT and images by Midjourney) + speech synthesis (online speech synthesis service https://play.ht)
  - Automatically conserve character consistency: 1) extract text description from input character design images using ViT and 2) given the text description, instruct ChatGPT to encode the character appearance in generating the prompts used for image generation.
- Class I overall project grade.
- For final report please see: https://github.com/MeixiChenTracy/MEng_project.

Improving Few-Shot Performance of Language Models via Nearest Neighbor Learning
(published as second author *arxiv:2212.02216*)

*Intern project, **Tencent** (Apr. 2022 - Aug. 2022)*

- Calibrated in-context learning by retrieving nearest neighbours from datastore (additional KNN classifier).
  - In-context learning concatenates few training examples in one prompt as demonstrations; it is highly sensitive to factors like ordering of samples, prompting templates and verbalizers.
  - KNN helps - correct decision boundary using similar support samples according to their top representations of pre-trained language model.
- Handle noise in retrieved neighbours by optionally training two small modules:
  - Adaptive neighbour selection: dynamically decide the number of neighbours to retrieve.
  - Feature regularisation: separate instances with similar representations but from different classes.
- Performance with and without training the noise handling modules:
  - Without parameter training: achieved superior results compared to no-training benchmarks, and comparable performance with some training-required benchmarks.
  - With parameter training to handle noise: performance boosted more.
  - For full result please see *arxiv:2212.02216.*

Natural language tags clustering

Intern project, Tencent (Jan. 2022 - Apr. 2022)

- Word2vec embeddings trained on game-related domain text resources from:
  - large game-related forum corpora (limited appearance of specific tags).
  - search engine results specifically for tag words.
- Clustered tags by their embeddings using K-Means and hierarchical clustering.
- Knowledge graph schema construction employed the output tags and clusters, saving manual effort.

Intent classification for dialogue system

Intern Project, CMB Network Technology (Jun. 2021 - Sept. 2021)

- Aimed to classify user intention in the online banking context. The boundaries of some classes were very close to each other, and ground truth classes were based on a set of complicated rules defined by humans.
- Modeled by: finBERT + modified triplet network which automatically generates more “hard” triplets.
- The best model achieved final accuracy above 98%, average recall and precision above 96%.

Other internships

DST Rental - Data Analysis Intern *(Oct. 2021 - Dec. 2021)*

- Predicted order renewals and early-terminatations.
- Modeled by:
  - binning continuous data by ChiMerge and equal-frequency binning,
  - encoding bins with WOE encoding,

  - modeling data with logic regression, neural network, and regression tree, respectively.
- Achieved total accuracy above 95%, 90% precision and recall on negative samples, 80% recall and 60% precision on positive samples (positive samples contribute to a very small amount of data).

Dexta Robotics - Unity Intern *(Jun. 2019 - Sept. 2019)*

- 3D bounding boxes automatic generation in Unity from input 3d model.
  - In Python: implemented Graphic calculations.
  - In C#: developed interactions with Unity.
- Attended the SIGGRAPH 2019 Summit as a representative of Dexta Robotics, and later visited companies in Silicon Valley such as Facebook, promoting our product and providing technical support.